

PHỤ LỤC

Phụ lục 1: Bảng giá trị của hàm $f(x) = \frac{1}{\sqrt{2\pi}} e^{\frac{-x^2}{2}}$

	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.3989	0.3989	0.3989	0.3988	0.3986	0.3984	0.3982	0.3980	0.3977	0.3973
0.1	0.3970	0.3965	0.3961	0.3956	0.3951	0.3945	0.3939	0.3932	0.3925	0.3918
0.2	0.3910	0.3902	0.3894	0.3885	0.3876	0.3867	0.3857	0.3847	0.3836	0.3825
0.3	0.3814	0.3802	0.3790	0.3778	0.3765	0.3752	0.3739	0.3725	0.3712	0.3697
0.4	0.3683	0.3668	0.3653	0.3637	0.3621	0.3605	0.3589	0.3572	0.3555	0.3538
0.5	0.3521	0.3503	0.3485	0.3467	0.3448	0.3429	0.3410	0.3391	0.3372	0.3352
0.6	0.3332	0.3312	0.3292	0.3271	0.3251	0.3230	0.3209	0.3187	0.3166	0.3144
0.7	0.3123	0.3101	0.3079	0.3056	0.3034	0.3011	0.2989	0.2966	0.2943	0.2920
0.8	0.2897	0.2874	0.2850	0.2827	0.2803	0.2780	0.2756	0.2732	0.2709	0.2685
0.9	0.2661	0.2637	0.2613	0.2589	0.2565	0.2541	0.2516	0.2492	0.2468	0.2444
1.0	0.2420	0.2396	0.2371	0.2347	0.2323	0.2299	0.2275	0.2251	0.2227	0.2203
1.1	0.2179	0.2155	0.2131	0.2107	0.2083	0.2059	0.2036	0.2012	0.1989	0.1965
1.2	0.1942	0.1919	0.1895	0.1872	0.1849	0.1826	0.1804	0.1781	0.1758	0.1736
1.3	0.1714	0.1691	0.1669	0.1647	0.1626	0.1604	0.1582	0.1561	0.1539	0.1518
1.4	0.1497	0.1476	0.1456	0.1435	0.1415	0.1394	0.1374	0.1354	0.1334	0.1315
1.5	0.1295	0.1276	0.1257	0.1238	0.1219	0.1200	0.1182	0.1163	0.1145	0.1127
1.6	0.1109	0.1092	0.1074	0.1057	0.1040	0.1023	0.1006	0.0989	0.0973	0.0957
1.7	0.0940	0.0925	0.0909	0.0893	0.0878	0.0863	0.0848	0.0833	0.0818	0.0804
1.8	0.0790	0.0775	0.0761	0.0748	0.0734	0.0721	0.0707	0.0694	0.0681	0.0669
1.9	0.0656	0.0644	0.0632	0.0620	0.0608	0.0596	0.0584	0.0573	0.0562	0.0551
2.0	0.0540	0.0529	0.0519	0.0508	0.0498	0.0488	0.0478	0.0468	0.0459	0.0449
2.1	0.0440	0.0431	0.0422	0.0413	0.0404	0.0396	0.0387	0.0379	0.0371	0.0363
2.2	0.0355	0.0347	0.0339	0.0332	0.0325	0.0317	0.0310	0.0303	0.0297	0.0290
2.3	0.0283	0.0277	0.0270	0.0264	0.0258	0.0252	0.0246	0.0241	0.0235	0.0229
2.4	0.0224	0.0219	0.0213	0.0208	0.0203	0.0198	0.0194	0.0189	0.0184	0.0180
2.5	0.0175	0.0171	0.0167	0.0163	0.0158	0.0154	0.0151	0.0147	0.0143	0.0139
2.6	0.0136	0.0132	0.0129	0.0126	0.0122	0.0119	0.0116	0.0113	0.0110	0.0107
2.7	0.0104	0.0101	0.0099	0.0096	0.0093	0.0091	0.0088	0.0086	0.0084	0.0081
2.8	0.0079	0.0077	0.0075	0.0073	0.0071	0.0069	0.0067	0.0065	0.0063	0.0061
2.9	0.0060	0.0058	0.0056	0.0055	0.0053	0.0051	0.0050	0.0048	0.0047	0.0046
3.0	0.0044	0.0043	0.0042	0.0040	0.0039	0.0038	0.0037	0.0036	0.0035	0.0034
3.1	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026	0.0025	0.0025
3.2	0.0024	0.0023	0.0022	0.0022	0.0021	0.0020	0.0020	0.0019	0.0018	0.0018
3.3	0.0017	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014	0.0013	0.0013
3.4	0.0012	0.0012	0.0012	0.0011	0.0011	0.0010	0.0010	0.0010	0.0009	0.0009
3.5	0.0009	0.0008	0.0008	0.0008	0.0008	0.0007	0.0007	0.0007	0.0007	0.0006
3.6	0.0006	0.0006	0.0006	0.0005	0.0005	0.0005	0.0005	0.0005	0.0005	0.0004
3.7	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0003	0.0003	0.0003	0.0003
3.8	0.0003	0.0003	0.0003	0.0003	0.0003	0.0002	0.0002	0.0002	0.0002	0.0002
3.9	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002	0.0001	0.0001

Phụ lục 2: Bảng giá trị của hàm $\varphi(x) = \frac{1}{\sqrt{2\pi}} \int_0^x e^{-\frac{t^2}{2}} dt$

	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.0000	0.0040	0.0080	0.0120	0.0160	0.0199	0.0239	0.0279	0.0319	0.0359
0.1	0.0398	0.0438	0.0478	0.0517	0.0557	0.0596	0.0636	0.0675	0.0714	0.0753
0.2	0.0793	0.0832	0.0871	0.0910	0.0948	0.0987	0.1026	0.1064	0.1103	0.1141
0.3	0.1179	0.1217	0.1255	0.1293	0.1331	0.1368	0.1406	0.1443	0.1480	0.1517
0.4	0.1554	0.1591	0.1628	0.1664	0.1700	0.1736	0.1772	0.1808	0.1844	0.1879
0.5	0.1915	0.1950	0.1985	0.2019	0.2054	0.2088	0.2123	0.2157	0.2190	0.2224
0.6	0.2257	0.2291	0.2324	0.2357	0.2389	0.2422	0.2454	0.2486	0.2517	0.2549
0.7	0.2580	0.2611	0.2642	0.2673	0.2704	0.2734	0.2764	0.2794	0.2823	0.2852
0.8	0.2881	0.2910	0.2939	0.2967	0.2995	0.3023	0.3051	0.3078	0.3106	0.3133
0.9	0.3159	0.3186	0.3212	0.3238	0.3264	0.3289	0.3315	0.3340	0.3365	0.3389
1.0	0.3413	0.3438	0.3461	0.3485	0.3508	0.3531	0.3554	0.3577	0.3599	0.3621
1.1	0.3643	0.3665	0.3686	0.3708	0.3729	0.3749	0.3770	0.3790	0.3810	0.3830
1.2	0.3849	0.3869	0.3888	0.3907	0.3925	0.3944	0.3962	0.3980	0.3997	0.4015
1.3	0.4032	0.4049	0.4066	0.4082	0.4099	0.4115	0.4131	0.4147	0.4162	0.4177
1.4	0.4192	0.4207	0.4222	0.4236	0.4251	0.4265	0.4279	0.4292	0.4306	0.4319
1.5	0.4332	0.4345	0.4357	0.4370	0.4382	0.4394	0.4406	0.4418	0.4429	0.4441
1.6	0.4452	0.4463	0.4474	0.4484	0.4495	0.4505	0.4515	0.4525	0.4535	0.4545
1.7	0.4554	0.4564	0.4573	0.4582	0.4591	0.4599	0.4608	0.4616	0.4625	0.4633
1.8	0.4641	0.4649	0.4656	0.4664	0.4671	0.4678	0.4686	0.4693	0.4699	0.4706
1.9	0.4713	0.4719	0.4726	0.4732	0.4738	0.4744	0.4750	0.4756	0.4761	0.4767
2.0	0.4772	0.4778	0.4783	0.4788	0.4793	0.4798	0.4803	0.4808	0.4812	0.4817
2.1	0.4821	0.4826	0.4830	0.4834	0.4838	0.4842	0.4846	0.4850	0.4854	0.4857
2.2	0.4861	0.4864	0.4868	0.4871	0.4875	0.4878	0.4881	0.4884	0.4887	0.4890
2.3	0.4893	0.4896	0.4898	0.4901	0.4904	0.4906	0.4909	0.4911	0.4913	0.4916
2.4	0.4918	0.4920	0.4922	0.4925	0.4927	0.4929	0.4931	0.4932	0.4934	0.4936
2.5	0.4938	0.4940	0.4941	0.4943	0.4945	0.4946	0.4948	0.4949	0.4951	0.4952
2.6	0.4953	0.4955	0.4956	0.4957	0.4959	0.4960	0.4961	0.4962	0.4963	0.4964
2.7	0.4965	0.4966	0.4967	0.4968	0.4969	0.4970	0.4971	0.4972	0.4973	0.4974
2.8	0.4974	0.4975	0.4976	0.4977	0.4977	0.4978	0.4979	0.4979	0.4980	0.4981
2.9	0.4981	0.4982	0.4982	0.4983	0.4984	0.4984	0.4985	0.4985	0.4986	0.4986
3.0	0.4987	0.4987	0.4987	0.4988	0.4988	0.4989	0.4989	0.4989	0.4990	0.4990
3.1	0.4990	0.4991	0.4991	0.4991	0.4992	0.4992	0.4992	0.4992	0.4993	0.4993
3.2	0.4993	0.4993	0.4994	0.4994	0.4994	0.4994	0.4994	0.4995	0.4995	0.4995
3.3	0.4995	0.4995	0.4995	0.4996	0.4996	0.4996	0.4996	0.4996	0.4996	0.4997
3.4	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4998
3.5	0.4998	0.4998	0.4998	0.4998	0.4998	0.4998	0.4998	0.4998	0.4998	0.4998
3.6	0.4998	0.4998	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999
3.7	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999
3.8	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999
3.9	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000

Phụ lục 3: Bảng giá trị phân vị chuẩn tắc $Z \sim N(0,1)$ với $P(Z < z_\alpha) = \alpha$

α	z_α	α	z_α	α	z_α	α	z_α	α	z_α
0.500	0.000	0.600	0.253	0.700	0.524	0.800	0.842	0.900	1.282
0.505	0.013	0.605	0.266	0.705	0.539	0.805	0.860	0.905	1.311
0.510	0.025	0.610	0.279	0.710	0.553	0.810	0.878	0.910	1.341
0.515	0.038	0.615	0.292	0.715	0.568	0.815	0.896	0.915	1.372
0.520	0.050	0.620	0.305	0.720	0.583	0.820	0.915	0.920	1.405
0.525	0.063	0.625	0.319	0.725	0.598	0.825	0.935	0.925	1.440
0.530	0.075	0.630	0.332	0.730	0.613	0.830	0.954	0.930	1.476
0.535	0.088	0.635	0.345	0.735	0.628	0.835	0.974	0.935	1.514
0.540	0.100	0.640	0.358	0.740	0.643	0.840	0.994	0.940	1.555
0.545	0.113	0.645	0.372	0.745	0.659	0.845	1.015	0.945	1.598
0.550	0.126	0.650	0.385	0.750	0.674	0.850	1.036	0.950	1.645
0.555	0.138	0.655	0.399	0.755	0.690	0.855	1.058	0.955	1.695
0.560	0.151	0.660	0.412	0.760	0.706	0.860	1.080	0.960	1.751
0.565	0.164	0.665	0.426	0.765	0.722	0.865	1.103	0.965	1.812
0.570	0.176	0.670	0.440	0.770	0.739	0.870	1.126	0.970	1.881
0.575	0.189	0.675	0.454	0.775	0.755	0.875	1.150	0.975	1.960
0.580	0.202	0.680	0.468	0.780	0.772	0.880	1.175	0.980	2.054
0.585	0.215	0.685	0.482	0.785	0.789	0.885	1.200	0.985	2.170
0.590	0.228	0.690	0.496	0.790	0.806	0.890	1.227	0.990	2.326
0.595	0.240	0.695	0.510	0.795	0.824	0.895	1.254	0.995	2.576

Phụ lục 4: Bảng giá trị phân vị student $T \sim t(n)$ với $P(T < t_\alpha) = \alpha$

df	0.900	0.905	0.910	0.915	0.920	0.925	0.930	0.935	0.940	0.945	0.950
1	3.078	3.251	3.442	3.655	3.895	4.165	4.474	4.829	5.242	5.730	6.314
2	1.886	1.953	2.026	2.104	2.189	2.282	2.383	2.495	2.620	2.760	2.920
3	1.638	1.688	1.741	1.798	1.859	1.924	1.995	2.072	2.156	2.249	2.353
4	1.533	1.577	1.623	1.671	1.723	1.778	1.838	1.902	1.971	2.048	2.132
5	1.476	1.516	1.558	1.602	1.649	1.699	1.753	1.810	1.873	1.941	2.015
6	1.440	1.478	1.517	1.559	1.603	1.650	1.700	1.754	1.812	1.874	1.943
7	1.415	1.451	1.489	1.529	1.572	1.617	1.664	1.715	1.770	1.830	1.895
8	1.397	1.432	1.469	1.508	1.549	1.592	1.638	1.687	1.740	1.797	1.860
9	1.383	1.418	1.454	1.492	1.532	1.574	1.619	1.666	1.718	1.773	1.833
10	1.372	1.406	1.442	1.479	1.518	1.559	1.603	1.650	1.700	1.754	1.812
11	1.363	1.397	1.432	1.468	1.507	1.548	1.591	1.636	1.686	1.738	1.796
12	1.356	1.389	1.424	1.460	1.498	1.538	1.580	1.626	1.674	1.726	1.782
13	1.350	1.383	1.417	1.453	1.490	1.530	1.572	1.616	1.664	1.715	1.771
14	1.345	1.377	1.411	1.447	1.484	1.523	1.565	1.609	1.656	1.706	1.761
15	1.341	1.373	1.406	1.441	1.478	1.517	1.558	1.602	1.649	1.699	1.753
16	1.337	1.369	1.402	1.437	1.474	1.512	1.553	1.596	1.642	1.692	1.746
17	1.333	1.365	1.398	1.433	1.469	1.508	1.548	1.591	1.637	1.686	1.740
18	1.330	1.362	1.395	1.429	1.466	1.504	1.544	1.587	1.632	1.681	1.734
19	1.328	1.359	1.392	1.426	1.462	1.500	1.540	1.583	1.628	1.677	1.729
20	1.325	1.357	1.389	1.424	1.459	1.497	1.537	1.579	1.624	1.672	1.725
21	1.323	1.354	1.387	1.421	1.457	1.494	1.534	1.576	1.621	1.669	1.721
22	1.321	1.352	1.385	1.419	1.454	1.492	1.531	1.573	1.618	1.665	1.717
23	1.319	1.350	1.383	1.417	1.452	1.489	1.529	1.570	1.615	1.662	1.714
24	1.318	1.349	1.381	1.415	1.450	1.487	1.526	1.568	1.612	1.660	1.711
25	1.316	1.347	1.379	1.413	1.448	1.485	1.524	1.566	1.610	1.657	1.708
26	1.315	1.346	1.378	1.411	1.446	1.483	1.522	1.564	1.608	1.655	1.706
27	1.314	1.344	1.376	1.410	1.445	1.482	1.521	1.562	1.606	1.653	1.703
28	1.313	1.343	1.375	1.408	1.443	1.480	1.519	1.560	1.604	1.651	1.701
29	1.311	1.342	1.374	1.407	1.442	1.479	1.517	1.558	1.602	1.649	1.699
30	1.310	1.341	1.373	1.406	1.441	1.477	1.516	1.557	1.600	1.647	1.697
40	1.303	1.333	1.365	1.397	1.432	1.468	1.506	1.546	1.589	1.635	1.684
50	1.299	1.329	1.360	1.392	1.426	1.462	1.500	1.539	1.582	1.627	1.676
60	1.296	1.326	1.357	1.389	1.423	1.458	1.496	1.535	1.577	1.622	1.671
70	1.294	1.323	1.354	1.386	1.420	1.456	1.493	1.532	1.574	1.619	1.667
80	1.292	1.322	1.353	1.385	1.418	1.453	1.491	1.530	1.572	1.616	1.664
90	1.291	1.321	1.351	1.383	1.417	1.452	1.489	1.528	1.570	1.614	1.662
100	1.290	1.320	1.350	1.382	1.416	1.451	1.488	1.527	1.568	1.613	1.660
200	1.286	1.315	1.345	1.377	1.410	1.445	1.482	1.520	1.561	1.605	1.653
300	1.284	1.314	1.344	1.376	1.409	1.443	1.480	1.518	1.559	1.603	1.650
400	1.284	1.313	1.343	1.375	1.408	1.442	1.479	1.517	1.558	1.602	1.649

Phụ lục 4: Bảng giá trị phân vị student $T \sim t(n)$ với $P(T < t_\alpha) = \alpha$ (tiếp theo)

df	0.955	0.960	0.965	0.970	0.975	0.980	0.985	0.990	0.995
1	7.026	7.916	9.058	10.579	12.706	15.895	21.205	31.821	63.657
2	3.104	3.320	3.578	3.896	4.303	4.849	5.643	6.965	9.925
3	2.471	2.605	2.763	2.951	3.182	3.482	3.896	4.541	5.841
4	2.226	2.333	2.456	2.601	2.776	2.999	3.298	3.747	4.604
5	2.098	2.191	2.297	2.422	2.571	2.757	3.003	3.365	4.032
6	2.019	2.104	2.201	2.313	2.447	2.612	2.829	3.143	3.707
7	1.966	2.046	2.136	2.241	2.365	2.517	2.715	2.998	3.499
8	1.928	2.004	2.090	2.189	2.306	2.449	2.634	2.896	3.355
9	1.899	1.973	2.055	2.150	2.262	2.398	2.574	2.821	3.250
10	1.877	1.948	2.028	2.120	2.228	2.359	2.527	2.764	3.169
11	1.859	1.928	2.007	2.096	2.201	2.328	2.491	2.718	3.106
12	1.844	1.912	1.989	2.076	2.179	2.303	2.461	2.681	3.055
13	1.832	1.899	1.974	2.060	2.160	2.282	2.436	2.650	3.012
14	1.821	1.887	1.962	2.046	2.145	2.264	2.415	2.624	2.977
15	1.812	1.878	1.951	2.034	2.131	2.249	2.397	2.602	2.947
16	1.805	1.869	1.942	2.024	2.120	2.235	2.382	2.583	2.921
17	1.798	1.862	1.934	2.015	2.110	2.224	2.368	2.567	2.898
18	1.792	1.855	1.926	2.007	2.101	2.214	2.356	2.552	2.878
19	1.786	1.850	1.920	2.000	2.093	2.205	2.346	2.539	2.861
20	1.782	1.844	1.914	1.994	2.086	2.197	2.336	2.528	2.845
21	1.777	1.840	1.909	1.988	2.080	2.189	2.328	2.518	2.831
22	1.773	1.835	1.905	1.983	2.074	2.183	2.320	2.508	2.819
23	1.770	1.832	1.900	1.978	2.069	2.177	2.313	2.500	2.807
24	1.767	1.828	1.896	1.974	2.064	2.172	2.307	2.492	2.797
25	1.764	1.825	1.893	1.970	2.060	2.167	2.301	2.485	2.787
26	1.761	1.822	1.890	1.967	2.056	2.162	2.296	2.479	2.779
27	1.758	1.819	1.887	1.963	2.052	2.158	2.291	2.473	2.771
28	1.756	1.817	1.884	1.960	2.048	2.154	2.286	2.467	2.763
29	1.754	1.814	1.881	1.957	2.045	2.150	2.282	2.462	2.756
30	1.752	1.812	1.879	1.955	2.042	2.147	2.278	2.457	2.750
40	1.737	1.796	1.862	1.936	2.021	2.123	2.250	2.423	2.704
50	1.729	1.787	1.852	1.924	2.009	2.109	2.234	2.403	2.678
60	1.723	1.781	1.845	1.917	2.000	2.099	2.223	2.390	2.660
70	1.719	1.776	1.840	1.912	1.994	2.093	2.215	2.381	2.648
80	1.716	1.773	1.836	1.908	1.990	2.088	2.209	2.374	2.639
90	1.714	1.771	1.834	1.905	1.987	2.084	2.205	2.368	2.632
100	1.712	1.769	1.832	1.902	1.984	2.081	2.201	2.364	2.626
200	1.704	1.760	1.822	1.892	1.972	2.067	2.186	2.345	2.601
300	1.701	1.757	1.818	1.888	1.968	2.063	2.180	2.339	2.592
400	1.700	1.755	1.817	1.886	1.966	2.060	2.178	2.336	2.588

Phụ lục 5: Bảng giá trị phân vị chi bình phương $\chi^2 \sim \chi^2(n)$ với $P(\chi^2 < \chi^2_\alpha) = \alpha$

df	0.050	0.045	0.040	0.035	0.030	0.025	0.020	0.015	0.010	0.005
1	0.004	0.003	0.003	0.002	0.001	0.001	0.001	0.000	0.000	0.000
2	0.103	0.092	0.082	0.071	0.061	0.051	0.040	0.030	0.020	0.010
3	0.352	0.326	0.300	0.273	0.245	0.216	0.185	0.152	0.115	0.072
4	0.711	0.670	0.627	0.582	0.535	0.484	0.429	0.368	0.297	0.207
5	1.145	1.090	1.031	0.969	0.903	0.831	0.752	0.662	0.554	0.412
6	1.635	1.566	1.492	1.414	1.330	1.237	1.134	1.016	0.872	0.676
7	2.167	2.085	1.997	1.903	1.802	1.690	1.564	1.418	1.239	0.989
8	2.733	2.638	2.537	2.428	2.310	2.180	2.032	1.860	1.646	1.344
9	3.325	3.218	3.105	2.982	2.848	2.700	2.532	2.335	2.088	1.735
10	3.940	3.822	3.697	3.561	3.412	3.247	3.059	2.837	2.558	2.156
11	4.575	4.446	4.309	4.160	3.997	3.816	3.609	3.363	3.053	2.603
12	5.226	5.087	4.939	4.778	4.601	4.404	4.178	3.910	3.571	3.074
13	5.892	5.743	5.584	5.411	5.221	5.009	4.765	4.476	4.107	3.565
14	6.571	6.412	6.243	6.058	5.856	5.629	5.368	5.057	4.660	4.075
15	7.261	7.094	6.914	6.718	6.503	6.262	5.985	5.653	5.229	4.601
16	7.962	7.785	7.596	7.390	7.163	6.908	6.614	6.263	5.812	5.142
17	8.672	8.487	8.288	8.071	7.832	7.564	7.255	6.884	6.408	5.697
18	9.390	9.197	8.989	8.762	8.512	8.231	7.906	7.516	7.015	6.265
19	10.117	9.915	9.698	9.462	9.200	8.907	8.567	8.159	7.633	6.844
20	10.851	10.641	10.415	10.169	9.897	9.591	9.237	8.810	8.260	7.434
21	11.591	11.374	11.140	10.884	10.601	10.283	9.915	9.471	8.897	8.034
22	12.338	12.113	11.870	11.605	11.313	10.982	10.600	10.139	9.542	8.643
23	13.091	12.858	12.607	12.333	12.030	11.689	11.293	10.815	10.196	9.260
24	13.848	13.609	13.350	13.067	12.754	12.401	11.992	11.497	10.856	9.886
25	14.611	14.365	14.098	13.807	13.484	13.120	12.697	12.187	11.524	10.520
26	15.379	15.125	14.851	14.551	14.219	13.844	13.409	12.882	12.198	11.160
27	16.151	15.891	15.609	15.301	14.959	14.573	14.125	13.583	12.879	11.808
28	16.928	16.660	16.371	16.055	15.704	15.308	14.847	14.290	13.565	12.461
29	17.708	17.434	17.138	16.813	16.454	16.047	15.574	15.002	14.256	13.121
30	18.493	18.212	17.908	17.576	17.208	16.791	16.306	15.719	14.953	13.787
40	26.509	26.168	25.799	25.394	24.944	24.433	23.838	23.113	22.164	20.707
50	34.764	34.370	33.943	33.473	32.951	32.357	31.664	30.818	29.707	27.991
60	43.188	42.746	42.266	41.738	41.150	40.482	39.699	38.744	37.485	35.534
70	51.739	51.253	50.724	50.143	49.495	48.758	47.893	46.836	45.442	43.275
80	60.391	59.864	59.290	58.659	57.955	57.153	56.213	55.061	53.540	51.172
90	69.126	68.560	67.944	67.266	66.509	65.647	64.635	63.394	61.754	59.196
100	77.929	77.326	76.671	75.949	75.142	74.222	73.142	71.818	70.065	67.328
200	168.279	167.380	166.400	165.320	164.111	162.728	161.100	159.096	156.432	152.241
300	260.878	259.752	258.524	257.169	255.650	253.912	251.864	249.338	245.972	240.663
400	354.641	353.324	351.886	350.299	348.520	346.482	344.078	341.112	337.155	330.903

Phụ lục 5: Bảng giá trị phân vị chi bình phương $\chi^2 \sim \chi^2(n)$ với $P(\chi^2 < \chi^2_\alpha) = \alpha$ (tiếp theo)

Df	0.950	0.955	0.960	0.965	0.970	0.975	0.980	0.985	0.990	0.995
1	3.841	4.019	4.218	4.445	4.709	5.024	5.412	5.916	6.635	7.879
2	5.991	6.202	6.438	6.705	7.013	7.378	7.824	8.399	9.210	10.597
3	7.815	8.049	8.311	8.607	8.947	9.348	9.837	10.465	11.345	12.838
4	9.488	9.742	10.026	10.345	10.712	11.143	11.668	12.339	13.277	14.860
5	11.070	11.342	11.644	11.985	12.375	12.833	13.388	14.098	15.086	16.750
6	12.592	12.879	13.198	13.557	13.968	14.449	15.033	15.777	16.812	18.548
7	14.067	14.369	14.703	15.079	15.509	16.013	16.622	17.398	18.475	20.278
8	15.507	15.822	16.171	16.563	17.010	17.535	18.168	18.974	20.090	21.955
9	16.919	17.246	17.608	18.015	18.480	19.023	19.679	20.513	21.666	23.589
10	18.307	18.646	19.021	19.442	19.922	20.483	21.161	22.021	23.209	25.188
11	19.675	20.025	20.412	20.846	21.342	21.920	22.618	23.503	24.725	26.757
12	21.026	21.386	21.785	22.232	22.742	23.337	24.054	24.963	26.217	28.300
13	22.362	22.733	23.142	23.601	24.125	24.736	25.472	26.403	27.688	29.819
14	23.685	24.065	24.485	24.956	25.493	26.119	26.873	27.827	29.141	31.319
15	24.996	25.385	25.816	26.298	26.848	27.488	28.259	29.235	30.578	32.801
16	26.296	26.695	27.136	27.629	28.191	28.845	29.633	30.629	32.000	34.267
17	27.587	27.995	28.445	28.949	29.523	30.191	30.995	32.011	33.409	35.718
18	28.869	29.285	29.745	30.259	30.845	31.526	32.346	33.382	34.805	37.156
19	30.144	30.568	31.037	31.561	32.158	32.852	33.687	34.742	36.191	38.582
20	31.410	31.843	32.321	32.855	33.462	34.170	35.020	36.093	37.566	39.997
21	32.671	33.111	33.597	34.141	34.759	35.479	36.343	37.434	38.932	41.401
22	33.924	34.373	34.867	35.420	36.049	36.781	37.659	38.768	40.289	42.796
23	35.172	35.628	36.131	36.693	37.332	38.076	38.968	40.094	41.638	44.181
24	36.415	36.878	37.389	37.960	38.609	39.364	40.270	41.413	42.980	45.559
25	37.652	38.123	38.642	39.221	39.880	40.646	41.566	42.725	44.314	46.928
26	38.885	39.363	39.889	40.477	41.146	41.923	42.856	44.031	45.642	48.290
27	40.113	40.598	41.132	41.729	42.407	43.195	44.140	45.331	46.963	49.645
28	41.337	41.828	42.370	42.975	43.662	44.461	45.419	46.626	48.278	50.993
29	42.557	43.055	43.604	44.217	44.913	45.722	46.693	47.915	49.588	52.336
30	43.773	44.277	44.834	45.455	46.160	46.979	47.962	49.199	50.892	53.672
40	55.758	56.324	56.946	57.640	58.428	59.342	60.436	61.812	63.691	66.766
50	67.505	68.123	68.804	69.563	70.423	71.420	72.613	74.111	76.154	79.490
60	79.082	79.749	80.482	81.299	82.225	83.298	84.580	86.188	88.379	91.952
70	90.531	91.242	92.024	92.895	93.881	95.023	96.388	98.098	100.425	104.215
80	101.879	102.632	103.459	104.380	105.422	106.629	108.069	109.874	112.329	116.321
90	113.145	113.936	114.806	115.774	116.869	118.136	119.648	121.542	124.116	128.299
100	124.342	125.170	126.079	127.092	128.237	129.561	131.142	133.120	135.807	140.169
200	233.994	235.118	236.351	237.722	239.270	241.058	243.187	245.845	249.445	255.264
300	341.395	342.746	344.228	345.873	347.731	349.874	352.425	355.605	359.906	366.844
400	447.632	449.175	450.866	452.744	454.862	457.305	460.211	463.832	468.724	476.606

Phụ lục 6: Bảng giá trị phân vị của phân phối Fisher ($\gamma = 1 - \alpha = 95\%$)

Df	1	2	3	4	5	6	7	8	9	10
1	161.448	199.500	215.707	224.583	230.162	233.986	236.768	238.883	240.543	241.882
2	18.513	19.000	19.164	19.247	19.296	19.330	19.353	19.371	19.385	19.396
3	10.128	9.552	9.277	9.117	9.013	8.941	8.887	8.845	8.812	8.786
4	7.709	6.944	6.591	6.388	6.256	6.163	6.094	6.041	5.999	5.964
5	6.608	5.786	5.409	5.192	5.050	4.950	4.876	4.818	4.772	4.735
6	5.987	5.143	4.757	4.534	4.387	4.284	4.207	4.147	4.099	4.060
7	5.591	4.737	4.347	4.120	3.972	3.866	3.787	3.726	3.677	3.637
8	5.318	4.459	4.066	3.838	3.687	3.581	3.500	3.438	3.388	3.347
9	5.117	4.256	3.863	3.633	3.482	3.374	3.293	3.230	3.179	3.137
10	4.965	4.103	3.708	3.478	3.326	3.217	3.135	3.072	3.020	2.978
11	4.844	3.982	3.587	3.357	3.204	3.095	3.012	2.948	2.896	2.854
12	4.747	3.885	3.490	3.259	3.106	2.996	2.913	2.849	2.796	2.753
13	4.667	3.806	3.411	3.179	3.025	2.915	2.832	2.767	2.714	2.671
14	4.600	3.739	3.344	3.112	2.958	2.848	2.764	2.699	2.646	2.602
15	4.543	3.682	3.287	3.056	2.901	2.790	2.707	2.641	2.588	2.544
16	4.494	3.634	3.239	3.007	2.852	2.741	2.657	2.591	2.538	2.494
17	4.451	3.592	3.197	2.965	2.810	2.699	2.614	2.548	2.494	2.450
18	4.414	3.555	3.160	2.928	2.773	2.661	2.577	2.510	2.456	2.412
19	4.381	3.522	3.127	2.895	2.740	2.628	2.544	2.477	2.423	2.378
20	4.351	3.493	3.098	2.866	2.711	2.599	2.514	2.447	2.393	2.348
21	4.325	3.467	3.072	2.840	2.685	2.573	2.488	2.420	2.366	2.321
22	4.301	3.443	3.049	2.817	2.661	2.549	2.464	2.397	2.342	2.297
23	4.279	3.422	3.028	2.796	2.640	2.528	2.442	2.375	2.320	2.275
24	4.260	3.403	3.009	2.776	2.621	2.508	2.423	2.355	2.300	2.255
25	4.242	3.385	2.991	2.759	2.603	2.490	2.405	2.337	2.282	2.236
26	4.225	3.369	2.975	2.743	2.587	2.474	2.388	2.321	2.265	2.220
27	4.210	3.354	2.960	2.728	2.572	2.459	2.373	2.305	2.250	2.204
28	4.196	3.340	2.947	2.714	2.558	2.445	2.359	2.291	2.236	2.190
29	4.183	3.328	2.934	2.701	2.545	2.432	2.346	2.278	2.223	2.177
30	4.171	3.316	2.922	2.690	2.534	2.421	2.334	2.266	2.211	2.165
40	4.085	3.232	2.839	2.606	2.449	2.336	2.249	2.180	2.124	2.077
50	4.034	3.183	2.790	2.557	2.400	2.286	2.199	2.130	2.073	2.026
60	4.001	3.150	2.758	2.525	2.368	2.254	2.167	2.097	2.040	1.993
70	3.978	3.128	2.736	2.503	2.346	2.231	2.143	2.074	2.017	1.969
80	3.960	3.111	2.719	2.486	2.329	2.214	2.126	2.056	1.999	1.951
90	3.947	3.098	2.706	2.473	2.316	2.201	2.113	2.043	1.986	1.938
100	3.936	3.087	2.696	2.463	2.305	2.191	2.103	2.032	1.975	1.927
200	3.888	3.041	2.650	2.417	2.259	2.144	2.056	1.985	1.927	1.878
300	3.873	3.026	2.635	2.402	2.244	2.129	2.040	1.969	1.911	1.862
400	3.865	3.018	2.627	2.394	2.237	2.121	2.032	1.962	1.903	1.854

Df	11	12	13	14	15	16	17	18	19	20
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1	242.983	243.906	244.690	245.364	245.950	246.464	246.918	247.323	247.686	248.013
2	19.405	19.413	19.419	19.424	19.429	19.433	19.437	19.440	19.443	19.446
3	8.763	8.745	8.729	8.715	8.703	8.692	8.683	8.675	8.667	8.660
4	5.936	5.912	5.891	5.873	5.858	5.844	5.832	5.821	5.811	5.803
5	4.704	4.678	4.655	4.636	4.619	4.604	4.590	4.579	4.568	4.558
6	4.027	4.000	3.976	3.956	3.938	3.922	3.908	3.896	3.884	3.874
7	3.603	3.575	3.550	3.529	3.511	3.494	3.480	3.467	3.455	3.445
8	3.313	3.284	3.259	3.237	3.218	3.202	3.187	3.173	3.161	3.150
9	3.102	3.073	3.048	3.025	3.006	2.989	2.974	2.960	2.948	2.936
10	2.943	2.913	2.887	2.865	2.845	2.828	2.812	2.798	2.785	2.774
11	2.818	2.788	2.761	2.739	2.719	2.701	2.685	2.671	2.658	2.646
12	2.717	2.687	2.660	2.637	2.617	2.599	2.583	2.568	2.555	2.544
13	2.635	2.604	2.577	2.554	2.533	2.515	2.499	2.484	2.471	2.459
14	2.565	2.534	2.507	2.484	2.463	2.445	2.428	2.413	2.400	2.388
15	2.507	2.475	2.448	2.424	2.403	2.385	2.368	2.353	2.340	2.328
16	2.456	2.425	2.397	2.373	2.352	2.333	2.317	2.302	2.288	2.276
17	2.413	2.381	2.353	2.329	2.308	2.289	2.272	2.257	2.243	2.230
18	2.374	2.342	2.314	2.290	2.269	2.250	2.233	2.217	2.203	2.191
19	2.340	2.308	2.280	2.256	2.234	2.215	2.198	2.182	2.168	2.155
20	2.310	2.278	2.250	2.225	2.203	2.184	2.167	2.151	2.137	2.124
21	2.283	2.250	2.222	2.197	2.176	2.156	2.139	2.123	2.109	2.096
22	2.259	2.226	2.198	2.173	2.151	2.131	2.114	2.098	2.084	2.071
23	2.236	2.204	2.175	2.150	2.128	2.109	2.091	2.075	2.061	2.048
24	2.216	2.183	2.155	2.130	2.108	2.088	2.070	2.054	2.040	2.027
25	2.198	2.165	2.136	2.111	2.089	2.069	2.051	2.035	2.021	2.007
26	2.181	2.148	2.119	2.094	2.072	2.052	2.034	2.018	2.003	1.990
27	2.166	2.132	2.103	2.078	2.056	2.036	2.018	2.002	1.987	1.974
28	2.151	2.118	2.089	2.064	2.041	2.021	2.003	1.987	1.972	1.959
29	2.138	2.104	2.075	2.050	2.027	2.007	1.989	1.973	1.958	1.945
30	2.126	2.092	2.063	2.037	2.015	1.995	1.976	1.960	1.945	1.932
40	2.038	2.003	1.974	1.948	1.924	1.904	1.885	1.868	1.853	1.839
50	1.986	1.952	1.921	1.895	1.871	1.850	1.831	1.814	1.798	1.784
60	1.952	1.917	1.887	1.860	1.836	1.815	1.796	1.778	1.763	1.748
70	1.928	1.893	1.863	1.836	1.812	1.790	1.771	1.753	1.737	1.722
80	1.910	1.875	1.845	1.817	1.793	1.772	1.752	1.734	1.718	1.703
90	1.897	1.861	1.830	1.803	1.779	1.757	1.737	1.720	1.703	1.688
100	1.886	1.850	1.819	1.792	1.768	1.746	1.726	1.708	1.691	1.676
200	1.837	1.801	1.769	1.742	1.717	1.694	1.674	1.656	1.639	1.623
300	1.821	1.785	1.753	1.725	1.700	1.677	1.657	1.638	1.621	1.606
400	1.813	1.776	1.745	1.717	1.691	1.669	1.648	1.630	1.613	1.597

df	21	22	23	24	25	26	27	28	29	30
1	248.309	248.579	248.826	249.052	249.260	249.453	249.631	249.797	249.951	250.095

2	19.448	19.450	19.452	19.454	19.456	19.457	19.459	19.460	19.461	19.462
3	8.654	8.648	8.643	8.639	8.634	8.630	8.626	8.623	8.620	8.617
4	5.795	5.787	5.781	5.774	5.769	5.763	5.759	5.754	5.750	5.746
5	4.549	4.541	4.534	4.527	4.521	4.515	4.510	4.505	4.500	4.496
6	3.865	3.856	3.849	3.841	3.835	3.829	3.823	3.818	3.813	3.808
7	3.435	3.426	3.418	3.410	3.404	3.397	3.391	3.386	3.381	3.376
8	3.140	3.131	3.123	3.115	3.108	3.102	3.095	3.090	3.084	3.079
9	2.926	2.917	2.908	2.900	2.893	2.886	2.880	2.874	2.869	2.864
10	2.764	2.754	2.745	2.737	2.730	2.723	2.716	2.710	2.705	2.700
11	2.636	2.626	2.617	2.609	2.601	2.594	2.588	2.582	2.576	2.570
12	2.533	2.523	2.514	2.505	2.498	2.491	2.484	2.478	2.472	2.466
13	2.448	2.438	2.429	2.420	2.412	2.405	2.398	2.392	2.386	2.380
14	2.377	2.367	2.357	2.349	2.341	2.333	2.326	2.320	2.314	2.308
15	2.316	2.306	2.297	2.288	2.280	2.272	2.265	2.259	2.253	2.247
16	2.264	2.254	2.244	2.235	2.227	2.220	2.212	2.206	2.200	2.194
17	2.219	2.208	2.199	2.190	2.181	2.174	2.167	2.160	2.154	2.148
18	2.179	2.168	2.159	2.150	2.141	2.134	2.126	2.119	2.113	2.107
19	2.144	2.133	2.123	2.114	2.106	2.098	2.090	2.084	2.077	2.071
20	2.112	2.102	2.092	2.082	2.074	2.066	2.059	2.052	2.045	2.039
21	2.084	2.073	2.063	2.054	2.045	2.037	2.030	2.023	2.016	2.010
22	2.059	2.048	2.038	2.028	2.020	2.012	2.004	1.997	1.990	1.984
23	2.036	2.025	2.014	2.005	1.996	1.988	1.981	1.973	1.967	1.961
24	2.015	2.003	1.993	1.984	1.975	1.967	1.959	1.952	1.945	1.939
25	1.995	1.984	1.974	1.964	1.955	1.947	1.939	1.932	1.926	1.919
26	1.978	1.966	1.956	1.946	1.938	1.929	1.921	1.914	1.907	1.901
27	1.961	1.950	1.940	1.930	1.921	1.913	1.905	1.898	1.891	1.884
28	1.946	1.935	1.924	1.915	1.906	1.897	1.889	1.882	1.875	1.869
29	1.932	1.921	1.910	1.901	1.891	1.883	1.875	1.868	1.861	1.854
30	1.919	1.908	1.897	1.887	1.878	1.870	1.862	1.854	1.847	1.841
40	1.826	1.814	1.803	1.793	1.783	1.775	1.766	1.759	1.751	1.744
50	1.771	1.759	1.748	1.737	1.727	1.718	1.710	1.702	1.694	1.687
60	1.735	1.722	1.711	1.700	1.690	1.681	1.672	1.664	1.656	1.649
70	1.709	1.696	1.685	1.674	1.664	1.654	1.646	1.637	1.629	1.622
80	1.689	1.677	1.665	1.654	1.644	1.634	1.626	1.617	1.609	1.602
90	1.675	1.662	1.650	1.639	1.629	1.619	1.610	1.601	1.593	1.586
100	1.663	1.650	1.638	1.627	1.616	1.607	1.598	1.589	1.581	1.573
200	1.609	1.596	1.583	1.572	1.561	1.551	1.542	1.533	1.524	1.516
300	1.591	1.578	1.565	1.554	1.543	1.533	1.523	1.514	1.505	1.497
400	1.582	1.569	1.556	1.545	1.534	1.523	1.514	1.505	1.496	1.488

	40	50	60	70	80	90	100	200	300	400
1	251.143	251.774	252.196	252.497	252.724	252.900	253.041	253.677	253.889	253.996
2	19.471	19.476	19.479	19.481	19.483	19.485	19.486	19.491	19.492	19.493

3	8.594	8.581	8.572	8.566	8.561	8.557	8.554	8.540	8.536	8.533
4	5.717	5.699	5.688	5.679	5.673	5.668	5.664	5.646	5.640	5.637
5	4.464	4.444	4.431	4.422	4.415	4.409	4.405	4.385	4.378	4.375
6	3.774	3.754	3.740	3.730	3.722	3.716	3.712	3.690	3.683	3.680
7	3.340	3.319	3.304	3.294	3.286	3.280	3.275	3.252	3.245	3.241
8	3.043	3.020	3.005	2.994	2.986	2.980	2.975	2.951	2.943	2.939
9	2.826	2.803	2.787	2.776	2.768	2.761	2.756	2.731	2.723	2.719
10	2.661	2.637	2.621	2.610	2.601	2.594	2.588	2.563	2.555	2.551
11	2.531	2.507	2.490	2.478	2.469	2.462	2.457	2.431	2.422	2.418
12	2.426	2.401	2.384	2.372	2.363	2.356	2.350	2.323	2.314	2.310
13	2.339	2.314	2.297	2.284	2.275	2.267	2.261	2.234	2.225	2.220
14	2.266	2.241	2.223	2.210	2.201	2.193	2.187	2.159	2.150	2.145
15	2.204	2.178	2.160	2.147	2.137	2.130	2.123	2.095	2.085	2.081
16	2.151	2.124	2.106	2.093	2.083	2.075	2.068	2.039	2.030	2.025
17	2.104	2.077	2.058	2.045	2.035	2.027	2.020	1.991	1.981	1.976
18	2.063	2.035	2.017	2.003	1.993	1.985	1.978	1.948	1.938	1.933
19	2.026	1.999	1.980	1.966	1.955	1.947	1.940	1.910	1.899	1.894
20	1.994	1.966	1.946	1.932	1.922	1.913	1.907	1.875	1.865	1.859
21	1.965	1.936	1.916	1.902	1.891	1.883	1.876	1.845	1.834	1.828
22	1.938	1.909	1.889	1.875	1.864	1.856	1.849	1.817	1.806	1.800
23	1.914	1.885	1.865	1.850	1.839	1.830	1.823	1.791	1.780	1.774
24	1.892	1.863	1.842	1.828	1.816	1.808	1.800	1.768	1.756	1.750
25	1.872	1.842	1.822	1.807	1.796	1.787	1.779	1.746	1.735	1.729
26	1.853	1.823	1.803	1.788	1.776	1.767	1.760	1.726	1.714	1.709
27	1.836	1.806	1.785	1.770	1.758	1.749	1.742	1.708	1.696	1.690
28	1.820	1.790	1.769	1.754	1.742	1.733	1.725	1.691	1.679	1.673
29	1.806	1.775	1.754	1.738	1.726	1.717	1.710	1.675	1.663	1.656
30	1.792	1.761	1.740	1.724	1.712	1.703	1.695	1.660	1.647	1.641
40	1.693	1.660	1.637	1.621	1.608	1.597	1.589	1.551	1.537	1.530
50	1.634	1.599	1.576	1.558	1.544	1.534	1.525	1.484	1.469	1.461
60	1.594	1.559	1.534	1.516	1.502	1.491	1.481	1.438	1.422	1.414
70	1.566	1.530	1.505	1.486	1.471	1.459	1.450	1.404	1.388	1.379
80	1.545	1.508	1.482	1.463	1.448	1.436	1.426	1.379	1.361	1.353
90	1.528	1.491	1.465	1.445	1.429	1.417	1.407	1.358	1.340	1.331
100	1.515	1.477	1.450	1.430	1.415	1.402	1.392	1.342	1.323	1.314
200	1.455	1.415	1.386	1.364	1.346	1.332	1.321	1.263	1.240	1.228
300	1.435	1.393	1.363	1.341	1.323	1.308	1.296	1.234	1.210	1.196
400	1.425	1.383	1.352	1.329	1.311	1.296	1.283	1.219	1.193	1.179